

Intermediation Reduces Punishment (and Reward)[†]

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This paper shows moral decision making is not well predicted by the overall fairness of an act but rather by the fairness of the consequences that follow directly. In laboratory experiments, third-party punishment for keeping money from a poorer player decreases when an intermediary actor is included in the transaction. This is true for completely passive intermediaries, even though intermediation decreases the payout of the poorest player and hurts equity, and because intermediation distances the transgressor from the outcome. A separate study shows rewards of charitable giving decrease when the saliency of an intermediary is increased. (JEL A13, D63, D64)

Each society learns to live with a certain amount of ... mis-behavior; but ... society must be able to marshal from within itself forces which will make as many of the faltering actors as possible revert to the behavior required for its proper functioning.

— Albert O. Hirschman (1970)

In August 2005, Merck sold the rights to a cancer drug to a small pharmaceutical company, Ovation, that immediately raised the price charged to patients by a factor of 7, and later by a factor of 10.¹ Firms often worry about negative reactions to price-gouging (e.g., Daniel Kahneman, Jack L. Knetsch, and Richard Thaler 1986b), but was Merck not equally concerned about consumer outrage in response to selling to a firm that might raise prices so dramatically?² This paper does not aim to fully explain behavior in this scenario, but rather to answer two provocative questions that it raises. First, is Merck avoiding blame simply by not doing the deed itself? That is, what if Merck was known to be solely responsible for and solely benefiting from the price increase? Could Merck possibly avoid some loss of goodwill merely because the price increase is not perceived to be a direct consequence of its actions? Second, if this is true, what would it teach us more generally about how we punish?

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[†]To comment on this article in the online discussion forum, or to view additional materials, visit the article page at <http://www.aeaweb.org/articles.php?doi=10.1257/mic.3.4.77>.

¹Berenson, Alex. 2006. "A Cancer Drug's Price Rise is Cause for Concern." *New York Times*. March 12, 2006, accessed August 18, 2011, <http://www.nytimes.com/2006/03/12/business/12price.html>.

²The small company, Ovation, raised prices on three different drugs in 2005 by at least a factor of 10 shortly after purchasing them.

This paper investigates these two questions of moral decision-making in the laboratory, where we have direct measures of punishment as well as increased control over objective measures of culpability. In the game, the first mover (Merck) has the option of being selfish at the expense of a poorer player (the patients) and whether to do this directly or through an intermediary agent (Ovation). An unaffected third party (i.e., the regulator, other customers) then has the opportunity to punish the first mover. The data show that the first mover is punished significantly less for keeping money when she “intermediates,” that is, when she intentionally adds another actor between her action and the outcome. Several experimental controls show that this decrease in punishment is not due to diffusion of responsibility, punisher confusion, lack of thought by the punisher, or merely the inclusion of a third party. Rather, we find that, punishment decreases because when she uses the intermediary, the first mover does not “directly interact” with the poorest player; it seems as though she has managed to distance herself from the bad outcome.

The experiment is also designed to pit the hypothesis that intermediation reduces punishment against predictions offered by economic models of fairness. In our experiment, using the intermediary can only harm equity and decrease the payout of the poorest player. Further, subjects are shown to believe intermediation will greatly hurt the poorest player and decrease equity. They are also shown to believe that others share this belief. Thus, one might predict that subjects would actually punish intermediation more. In other studies, subjects have been shown to have a preference for both equating outcomes (Ernst Fehr and Klaus Schmidt 1999) and maximizing the minimum payoff (Gary Charness and Matthew Rabin 2002) and are willing to punish deviations from these norms (Fehr and Urs Fischbacher 2004; Jeffrey Carpenter and Peter Hans Matthews 2009). Since intermediation is a deviation from both norms in our game, the punishment decisions run counter to these findings. We do not suggest these norms are absent in our experiment. These norms have been shown to be significant across a variety of contexts, and we have no reason to suspect they are not present here. Further, if they exist in our experiment, they act as a countervailing force. Hence when “intermediation” prevails as the better predictor of punishment, this means two things. First, “intermediation” has prevailed over a known, significant benchmark, indicating it may be first order in punishment decisions in such contexts. Second, though this paper is not reporting levels, the size of the effect may be larger than the reported estimates given the likely significant countervailing forces.

Moreover, since pro-social norms are not as strongly enforced with intermediation, socially preferable outcomes occur less frequently; the poorest player is dramatically worse off, making 36 percent as much as her counterpart in the same game without intermediaries. When the mere availability of indirectness is introduced to the environment, punishment becomes almost entirely futile, failing to sustain pro-social behavior. Additionally, when intermediaries are not only available but used, the poorest player makes less than 5 percent as much as the recipient does in the game without intermediaries. Though this comparison is subject to selection, it helps illustrate the potential antisocial consequences of intermediation reducing punishment.

In a follow-up study, subjects’ behavior in a charity and reward context is consistent with overweighting direct consequences. The experiment varies the saliency of

the fact that charitable donations pass through an intermediary before reaching the recipient. Increasing the saliency of the intermediary in descriptions given to unaffected third parties considerably decreased rewards they gave to donors. Together, the results suggest that indirectness of behavior attenuates both punishment or reward, and we hypothesize that this is perhaps so because indirectness decreases the total perceived “badness” or “goodness” of an act.

There exists a large literature in both Economics and Psychology pertaining to fairness generally as well as fairness with multiple parties. Some papers will be mentioned in the behavioral predictions section, but we will review this literature more extensively after the results, in Section IV. This structure will allow for a discussion of the literature in relation to the current results.

The study of intermediation is important because of its ubiquity. Intermediations such as Merck’s sale of its cancer drug are common. Many companies may face less backlash for outsourcing production to other firms who cut costs through questionable labor practices or by allowing more pollution than they would face if they took the same actions themselves. For example, many companies have their products produced in factories in China that use a dirty coal technology.³ One could imagine a far greater public reaction if these companies themselves owned the factories that were emitting the CO₂ into the atmosphere. Similarly, firms can avoid being labeled a “patent troll” if they sell a patent to another firm willing to sue the infringing party.⁴ Intermediation can be, and often is, less obvious than supply chains, selling, or outsourcing. Fehr, Oliver Hart, and Christian Zehnder (2011) show that experimental “suppliers” shirk less often when paid a low price if the “buyer” defers price determination to a market process rather than choosing a low price directly. This is true even though the market has a known low outcome the subjects have experienced many times. There are also many real world examples of firms having a staffing company or an industrial psychologist lay off workers, hoping perhaps that the employees might harbor less ill will towards the firm if employers do not deliver the pink slip themselves.

Intermediation is not restricted to firm and worker relationships. In divorce proceedings, it would be inappropriate if one called his soon-to-be ex wife a terrible mother (especially in front of one’s children), but maybe this impropriety is attenuated if one has his lawyer say it on his behalf. Like all of these examples, the experiment within will consider human intermediaries, though perhaps the concept can be more generalizable. Giving a friend cash for her wedding might be considered uncouth, but if the money is given through an online gift registry (by purchasing the item she has already selected), this might be viewed as more palatable.

The paper proceeds as follows. Section I details the laboratory experimental designs. Section II discusses predictions for the experiment from leading models of fairness. Section III describes the results from the laboratory experiments. Section IV explains the design as well as the results of the follow-up study involving

³Spencer, Jane. 2007. “Why China Could Blame its CO₂ on the West.” *Wall Street Journal*. November 12, 2007, accessed August 18, 2011, <http://online.wsj.com/article/SB119482231216689376.html>.

⁴Elinson, Zusha. 2009. “Intellectual Ventures Takes an Indirect Route to Court.” *The Recorder*. September 1, 2009, accessed August 18, 2011, <http://www.law.com/jsp/ca/PubArticleCA.jsp?id=1202433488815&slreturn=1&hblogin=1>.

charitable actions and rewards. Section V reviews related literature and discusses how our results relate. Finally, Section VI provides concluding remarks.

I. Experimental Design—*The Intermediation Game*

The Intermediation Game has four players—first mover, intermediary, receiver, and punisher (exact instructions as given to participants, along with slideshow of screen shots shown to participants, for every treatment is in the online Appendix). The first mover owns a dictator game (henceforth “DG”) worth \$10 to be played with the receiver. She has two options of what to do with the DG. She can play the game herself, or she can “sell” it to the intermediary. If she chooses to sell, she also chooses the price the intermediary must pay her for the DG. The first mover’s strategy space is the same whether she is choosing how much to keep in the DG or the price for which she will sell it: \$5, \$6, \$7, \$8, \$9, \$10. If the DG is sold to the intermediary, she plays it with the same receiver, and she must keep at least as much as she was forced to pay; she cannot lose money. The intermediary has a \$5 endowment, equal to half the size of the DG. Thus, holding fixed the rents extracted by the first mover, using the intermediary weakly reduces equity and weakly makes the poorest player (the receiver) worse off.

Finally, the punisher can then reduce the payoff of the first mover (and only the first mover).⁵ The punisher will make punishment decisions with information on the amount sent in the DG, whether the DG was sold, and for how much the DG was sold. The punisher can costlessly reduce the first mover’s payout to any non-negative amount. Punishment decisions are elicited via the strategy method: The punisher gives a decision for all possible outcomes of the game the other three are playing, and her decision for the scenario that is realized is enacted. The use of the strategy method is discussed below. The punisher is endowed with \$5. Thus, the intermediary and punisher both begin with \$5, and the first mover is deciding how to split \$10, so a 4-way, \$5 equal split is clearly feasible and can be guaranteed by the first mover if she does not sell the DG.

The intermediary has no say in the selling or pricing decisions, so the experiment avoids any complicity of the intermediary in the first mover’s rent extraction. Punishment was costless in order to increase power. The task of the punisher is akin to the task of a judge. Judges are not assessed costs for penalties they mete out; the effect their decisions have on others are incentive for truth-telling and thoughtful decision-making. The data do not reveal any patterns of careless decision-making (e.g., punishment is strictly increasing in the amount kept in the DG, as in Fehr and Fischbacher (2004) who use a costly punishment technology), so this assumption seems valid.⁶ The strategy method was utilized, so to minimize the number of scenarios posed to the punisher, all transactions in the game were in multiples of \$1 (sell prices and DG allocations). This reduced the number of scenarios to 27. The

⁵ If any potential punishment of the intermediary is redirected to the first mover, this will work against Result 1, but regardless will be controlled for in the data analysis.

⁶ In pilot experiments, punishment was costly. There was less punishment overall, but the punishment patterns described in Section III were the same.

punisher was thus asked for 27 punishment decisions, one at a time, in a random order. The order was reversed each subsequent session. The same random order was used in other treatments (as well as the reversal if there were two sessions). In each session, the game was repeated four times, with each subject playing each role once. The use of role-rematching is discussed in the next paragraph. They were informed that they were rematched into new groups in each period.^{7,8} They were paid for their earnings in one randomly chosen period. This is to discourage thinking of the game as a four period meta-game but rather focus on maximizing each period. To minimize learning about others' behavior, no feedback was given until all four periods were completed.

Strategy method and role rematching were integral in gathering useful data in a cost-minimizing manner. Both methods have been shown to alter reported level effects of behavior (see Jordi Brandts and Charness (2011) survey on strategy method and James Andreoni and Rachel Croson (2008) on rematching in public goods games). Our results, however, only rely on the assumption that these protocols have no consistent, significant interaction effects with the different treatments. There is no evidence for, or plausible model for, interaction effects of this sort for our context. In a recent survey, Brandts and Charness (2011) find no effect of using the strategy method on treatment effects in punishment experiments. Likewise, across a wide variety of social preference games, some that allow for reciprocity and punishment, Charness and Rabin (2005), "see no evidence that people make different choices when role reversal is used in the experimental design."

After the four periods, but before feedback was given, subjects were asked for their beliefs of how much the intermediary kept for each transaction price. They were paid \$0.50 if they were within \$0.50 of the average of how intermediaries actually behaved in that session (and \$0.50 if this scenario did not occur in this session).⁹ In the final session of *The Intermediation Game*, they were then asked to guess the beliefs of others. They were paid \$0.50 if they were within \$0.50 of the average guess of everyone in the room from the questions previously answered. After these third party, second order beliefs were obtained, feedback was given.

All experimental sessions were performed at the Computer Lab for Experimental Research (CLER) at Harvard Business School. All sessions consisted of 12, 16, 20, or 24 study participants recruited via e-mail from the Boston area. All subjects were under the age of 30.¹⁰ Per CLER guidelines, they were paid a \$10 show up fee plus their earnings from the experiment, which averaged \$4.57 for the game and \$3.37 for the belief elicitations. The experiment was conducted at computer terminals

⁷ An analysis of order effects and within-subject consistency can be found in the online Appendix.

⁸ Given sessions as small as 12 subjects, we could not eliminate being in the same group as another subject twice. Subjects were never in the same group as another subject in consecutive periods. When they were in the same group twice, the subject did not know with whom or when.

⁹ This incentivizes subjects to report the midpoint of the \$1 interval they believe to be most likely for their session. Beliefs were elicited in this way in order to make it simple and understandable for the subject.

¹⁰ In one session, the recruitment software experienced a bug and allowed subjects over 30 to sign up. These data are not included in the analysis. The main results do not change with their inclusion. In sum, over-30s punished more, punished intermediation less, and exhibited more outcome bias. The over-30s largely came from an unemployed and/or homeless population. They were confused by the instructions and were unfamiliar with computers, and consequently they did not finish in one hour and were let go.

using z-Tree 2.1.4 (Fischbacher 2007). No session took longer than one hour. Sixty four subjects participated in this treatment in August and September of 2008.

In all treatments, subjects were randomly assigned a computer upon entering the laboratory. After signing consent forms, experimental instructions were distributed. These were read aloud by the experimenter while the subjects followed along. The instructions were accompanied by a slideshow with screen shots from the experiment they were about to play (the slideshow and accompanying script for every treatment can be found in the online Appendix). Questions were done one-on-one. Only clarifying questions were answered. Subjects who asked strategic or normative questions were told "I cannot answer that" by the experimenter. Once there were no more questions, which never took more than 5 minutes, the game began.

Primarily to test the context-specificity of the results, a study in a charity-reward domain was also employed. The methodology is quite different, so to avoid confusion, its design and results follow in Section IV.

A. Control—No Intermediary Treatment

To evaluate the welfare effects due to the presence of intermediaries, a control experiment, with no intermediary, was also run. Thus the control was an instantiation of the *Third Party Punishment Game* (TP-DG, Fehr and Fischbacher 2004). Otherwise, everything was consistent with earlier sessions: punishment was costless, the punisher made \$5, and the DG was worth \$10. Throughout, this will be referred to as the *No Intermediary* treatment. This treatment was run in October 2008 and had 24 subjects.

B. Two Punishments Treatment

To understand better how subjects allocate blame, we added a treatment that was identical to *The Intermediation Game* but allowed the punisher to punish *both* the first mover and the intermediary. The punishment mechanism does not change: punishment is costless, and the punisher cannot reduce either player's payoff to negative amounts. She can punish both, one, or neither. The amount she chooses to punish one does not affect her strategy space for the other punishment decision. The punishments are again elicited using the strategy method, and though the scenarios are presented one at a time, the punishment decision for the first mover and the intermediary are elicited simultaneously. This treatment was run over two sessions in March 2009 with 20 and 16 subjects in the two sessions.

C. The Reflection Treatment

To encourage subjects to think about and see if they understood the game and particularly the strategies of other players, they were asked in this treatment to think about the game and write down their thoughts beforehand. This treatment was exactly the same as *The Intermediation Game* except subjects spent four minutes before the game writing on four blank sheets of paper for a minute apiece reflecting on four questions:

- (i) Why might the first mover sell the game?

- (ii) Why might the first mover not sell the game?
- (iii) What will happen if the first mover does not sell the game?
- (iv) What will happen if the first mover sells the game?

The questions are intentionally neutral and balanced across scenarios of selling and not selling. This treatment was to encourage the subjects to reflect about what is going on in the game and to test if the results hold for subjects who are clearly aware of the potentially dubious motives of other players. As Neeru Paharia et al. (2009) suggest, moral judgments seem to be intuitive, and when subjects are forced to reflect on motives, judgments can change significantly. This session was run in October 2008 and had 24 subjects.

D. *The Allow-Taking Game*

This game reframes *The Intermediation Game* to test the hypothesis that intermediation reduces punishment precisely because when the intermediary is used, the first mover is not directly interacting with the receiver. In this treatment, the first mover first plays a \$10 DG with the receiver, keeping \$5, \$6, \$7, \$8, \$9, or \$10 for herself. She then decides whether to allow the intermediary to take from the money the first mover sent the receiver in the DG.¹¹ If she allows the intermediary to take, the intermediary can take any non-negative integer amount up to the amount that the receiver was sent in the DG. The punishment technology and elicitation is identical to those in *The Intermediation Game*. Thus, *The Allow-Taking Game* is a reframing of *The Intermediation Game*, except the first mover directly interacts with the receiver whether or not the intermediary is included in the game. This session was run in October 2009 and had 24 subjects.

E. *The Forced-Taking Game*

This game is a reframing of *The Intermediation Game* (and hence also *The Allow-Taking Game*) designed to test two potential explanations why using an intermediary may reduce punishment in *The Intermediation Game* but not in *The Allow-Taking Game*. First is the hypothesis that selling the DG to the intermediary is less reprehensible in *The Intermediation Game* because it is framed as a market transaction (since the term “sell” is used). Second is the hypothesis that *The Allow-Taking Game* finds no difference in punishment because the act of taking is more outrageous than keeping, and changing this frame may change the punishment imposed. In *The Forced-Taking Game*, the first mover first decides who starts with the \$10, herself or the receiver. If she starts with the \$10, then she plays a \$10 DG with the receiver. If the receiver starts with the \$10, the first mover decides how much the intermediary

¹¹ To see if either switching from a giving frame to a taking frame, or switching from a market transaction frame (selling the DG) to a transfer frame drives the results of this game, see the design and results for *The Who's-The-Dictator Game* in the Appendix, sections A and B.

must take from the receiver and pass to her. The intermediary must take at least this much from the receiver's \$10, though she may take more. Anything she takes on top of this amount is the intermediary's profit for the game. Anything she does not take is the receiver's payout. After either process, the punisher can reduce the first mover's payout. The punishment technology and elicitation is identical to those in *The Intermediation Game*. Two sessions, with 20 and 24 subjects, were run in October 2009 at the Computer Lab for Experimental Research at Harvard Business School. Each session lasted one hour.

II. Behavioral Predictions

There is a class of models that we can extend to make predictions for *The Intermediation Game*. We will show that none will predict less punishment for rents extracted through the intermediary. Moreover, the most recent models will predict that punishment will be *greater* for money made via intermediation. That is, the public might punish Merck *more* for selling the drug to an unpunishable intermediary.

Most of the models discussed below were designed to speak directly to the context of our experiment. They were designed for two-person set-ups where fairness judgments come from a second person point-of-view. We can only consider reasonable extensions to our context and discuss what predictions we might get with the same intuition.

The simplest way to separate the competing theories is to ask what punishment levels they predict when the vector of payoff outcomes is the same, but the path taken is different. Let us consider the following two examples: First, the first mover keeps \$ X in the DG, and second, she sells the DG for \$ X and the intermediary then keeps \$ X , making zero profit. In both cases, the resulting payout is (\$ X , \$5, \$10 – \$ X , \$5) for the first mover, the intermediary, the receiver, and the punisher, respectively (recall the intermediary and the punisher are endowed with \$5 each). We will consider what punishments each class of models would predict for each path. We will use “Direct” to refer to the path where the first mover does not sell the DG and plays the DG herself, and we will use “Indirect” to refer to the path where the first mover sells the DG.

Outcome-Based Models of Fairness.—If punishers judge fairness based purely on the outcome, then punishments will be independent of whether the intermediary was used; punishment will be identical in the two cases raised above. George F. Loewenstein, Leigh Thompson, and Max H. Bazerman (1989) experimentally identify a utility function defined by a distaste for inequity over payouts, more so for disadvantaged payouts. This was later modeled by Fehr and Schmidt (1999) as well as Gary E. Bolton and Axel Ockenfels (2000) (to see a formalization of the behavioral predictions for *The Intermediation Game* from the model of Fehr and Schmidt 1999, please see section A in the Appendix). These models are only intended to provide understanding of preferences over outcomes, not *how* those outcomes came about. For that, many authors have added intentions to the model of fairness.

Intentions-Based Models of Fairness.—Rabin (1993) and Martin Dufwenberg and Georg Kirchsteiger (2004) define fairness through intent, and the relevant piece of intent for *The Intermediation Game* is beliefs.¹² In Rabin (1993), if player i believes player j believes player i will play a_i , and player j plays b_j , then player i judges the fairness of player j based on what profits would result from a_i and b_j regardless of what player i actually plays. The intuition is that holding fixed player i 's action at a_i , this creates a dictator game for player j over all the outcomes that are possible given a_i . Let us extend this intuition to the *Intermediation Game*. We are interested in punishment of the first mover, so consider the punisher's beliefs of the first mover's beliefs of others' actions. We will take those beliefs as given and consider what outcome she chooses given the space of outcomes the first mover is now choosing within.

Suppose the first mover makes $\$X$. If she does so "directly," there are no other actions or beliefs to consider; she has chosen the outcome $(\$X, \$5, \$10 - \$X, \$5)$. If she makes the $\$X$ "indirectly," then we have to ask—"what does the punisher believe the first mover believes the intermediary will do when sold the DG for $\$X$?" Say the punisher believes the first mover believes the intermediary will keep $\hat{Y}$, where $\hat{Y} \geq X$ and $\hat{Y} \leq 10$. Then, under our extended notion of intentions, the punisher will judge the first mover for choosing the allocation $(\$X, \$5 + (\hat{Y} - \$X), \$10 - \hat{Y}, \$5)$ regardless of the actual actions of the intermediary. Thus, in comparing this action (selling the DG for $\$X$) to not selling the DG and keeping $\$X$ directly, the intentions-based punisher judges the fairness of choosing $(\$X, \$5, \$10 - \$X, \$5)$ (no intermediation) versus $(\$X, \$5 + (\hat{Y} - \$X), \$10 - \hat{Y}, \$5)$ (intermediation). Notice that the two allocations differ only by a $\$(\hat{Y} - X)$ transfer from the receiver to the intermediary, a transfer from the poorest player to a player already endowed with the average pre-punishment payout ($\$5$). The transfer does not affect the payout of the punisher. In the main text, Rabin (1993) and Dufwenberg and Kirchsteiger (2004) do not define preferences over *others'* payouts. However, in this model, if a punisher is indifferent over who receives the $\$(Y - X)$, punishment is expected to be identical whether the first mover keeps $\$X$ in the DG or via intermediation. If punishers are not indifferent about others' payoffs, these intentions-based models are also outcome-based.

Intentions and Outcome-Based Models of Fairness.—Their experimental evidence leads Charness (2004) and Armin Falk, Fehr, and Fischbacher (2008) to conclude that models based on both intentions and outcomes outperform models based solely on either intentions or outcomes in predicting retributive behavior in standard laboratory games.¹³ We will now consider a class of models that combine preferences over the outcomes of others with the intentions-based framework from the previous subsection.

In a zero-sum game, the first leading candidate for preferences over others' outcomes is a preference for maximizing the minimum. Charness and Rabin (2002),

¹² The other piece of intent is the strategy space from which the judged action was chosen, but the strategy space is held constant in our game, so this will not be a part of the discussion.

¹³ Falk and Fischbacher (2006) provide one such unifying framework. However, agents in their model only have preferences over outcomes relative to their own.

Dirk Engelmann and Martin Strobel (2004), and Lars-Olof Johansson and Henrik Svedsäter (2009) all experimentally show that subjects like to maximize the minimum payoff among players.¹⁴ The second candidate is “inequity aversion,” as in Loewenstein, Thompson, and Bazerman (1989), Fehr and Schmidt (1999), and Gary E. Bolton and Axel Ockenfels (2000). We will consider both preferences coupled with intentions.

From the previous section, recall a punisher who considered the *intentions* of the first mover would judge her for choosing $(\$X, \$5, \$10 - \$X, \$5)$ when intermediating and $(\$X, \$5 + (\hat{Y} - \$X), \$ (10 - \hat{Y}), \$5)$ when not intermediating. In Appendix Section A, we compare the fairness of these outcomes with models of maximizing the minimum and inequity aversion. Not surprisingly, since intermediation weakly hurts the poorest player as well as equity, both models will predict weakly more punishment for intermediation, and strictly more when $\hat{Y} > 0$.

Diffusion of Responsibility.—Charness (2000) and Björn Bartling and Fischbacher (2008) both experimentally demonstrate that including a second, human agent who is partly responsible for the outcome will partially absolve the initial agent. In Bartling and Fischbacher’s (forthcoming) model of this concept, an agent’s responsibility is determined by her share of the total increase in the probability of the bad outcome occurring due to human agents; thus, if a second party increases the likelihood of the outcome, the principal’s responsibility decreases for the same bad outcome, holding fixed her actions. However, in our game, even when the first mover sells the game for $\$X$, she has, at the very least, guaranteed the receiver cannot be paid more than $\$10 - X$; thus, she is fully responsible for the second mover not making more than $\$10 - X$ as she would be had she kept $\$X$ in the DG. Further, by putting the DG in the hands of an unpunishable party, the probability of a worse outcome for the receiver weakly increases, so doing so, in fact, may make the first mover weakly *more* responsible than she would be had she not sold the game, and we would expect weakly more punishment. In addition, this model makes an even clearer prediction when the intermediary is perfectly altruistic. When the second agent does not transgress, she is not responsible for the outcome; thus, responsibility is not diffused, and punishment should not decrease. It thus seems reasonable that in any case, we should expect weakly more punishment for intermediating.

III. Results

The results in this section confirm the main hypothesis: Intermediation decreases punishment. This is not due to diffusion of responsibility, punisher confusion, lack of thought by the punisher, or merely the inclusion of a third party. Rather, the results suggest that intermediation reduces punishment because it allows the first mover to avoid directly interacting with the poorest player. Moreover, first movers frequently use the intermediary. This shifts money from the poorest player to the

¹⁴Though efficiency is also an important parameter (see Charness and Rabin 2002; Engelmann and Strobel 2004), all moves in the pre-punishment stage are zero-sum.

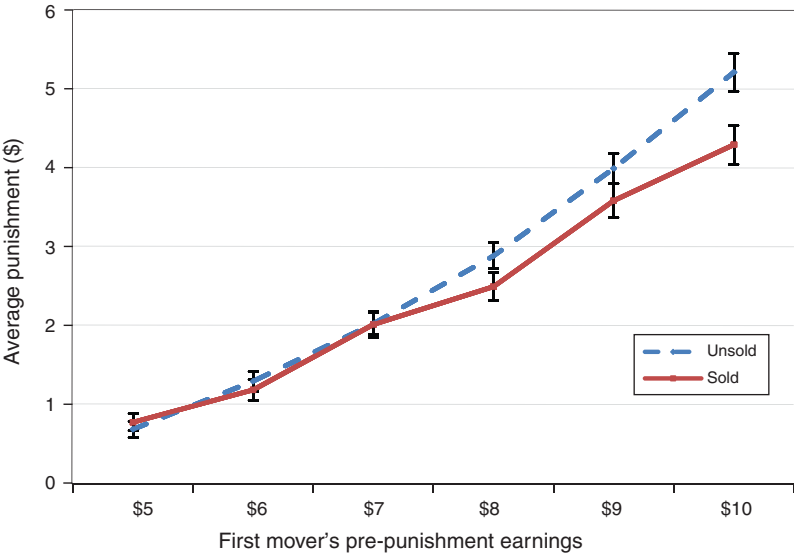


FIGURE 1. AVERAGE PUNISHMENTS FOR WHETHER THE DG WAS SOLD OR UNSOLD

endowed intermediary. As a result, the poorest player is considerably worse off due to the presence of the intermediary.

In *The Intermediation Game*, each punisher was asked to make a punishment decision for the same 27 scenarios (Table 1 of the online Appendix has a complete list of scenarios and average punishments) facilitating matched, nonparametric analysis of the data, which will be employed alongside OLS regression analyses. Unless otherwise noted in this section, reported results are from *The Intermediation Game* treatment.

Table A1, in the Appendix, provides summary statistics for a general understanding of how the game was played. Eighty-one percent of subjects punish in at least one scenario, and when they do punish, it is economically substantial, almost equal to average profit and fully 58 percent of the first mover's pre-punishment payout (see Table 7 in the online Appendix for more details on what may have determined the size of punishments). Figure 1 also shows just how large average punishments were in the experiment, with an average punishment over \$5 when the first mover keeps \$10 in the DG herself. Also note that 52 percent of subjects use the intermediary, so it is not an irrelevant alternative. Result 1 may help explain why.

RESULT 1:

Punishment significantly decreases when high levels of selfishness are done through intermediation rather than directly.

Figure 1 and Table 1 show that when the first mover keeps more than \$7, directness is punished more harshly: Average punishment is greater when the first mover extracts rents directly, and many subjects punish in this direction. At these profit levels, at least 2.5 times as many subjects punish direct actions more harshly than subjects who punish indirect actions more harshly. Using nonparametric two-tailed

TABLE 1—DIRECTNESS IS PUNISHED MORE HARSHLY^a

Profit of first mover	Avg. punishment		# Subjects who punish		<i>p</i> -values	
	DG unsold	DG sold	Unsold more	Sold more	Matched pair, signed rank ^b	Permutation test ^c
\$10	\$5.22	\$4.30	24	7	< 0.01	0.01
\$9	\$4.00	\$3.52	15	6	0.05	0.07
\$8	\$2.89	\$2.48	20	8	0.02	0.07
\$7	\$2.03	\$1.98	16	12	0.58	1.00
\$6	\$1.30	\$1.41	11	6	0.22	0.27
\$5	\$0.69	\$0.78	2	1	0.55	0.50

Notes: Pairs matched by subject. *p*-values are from two-tailed tests.

^aTo hold outcomes constant, only scenarios where intermediary makes no money included.

^bMatched pair signed rank test based on null hypothesis that punishment distributions are identical for direct and indirect action. Pairs matched by subject.

^cPermutation *p*-values based on 200,000 simulations per test.

tests—both a Wilcoxon signed rank test and a Fisher Pitman matched-pair permutation test—the punishment distributions are shown to be significantly different for each of these high profit levels of the first mover (See the last two columns of Table 1). In other words, when punishment (and misbehavior) is high, direct actions are frequently and significantly punished more.

The strongest case in support of Result 1 is in the first row of Table 1: When the first mover keeps *all* \$10 for herself, she is punished less when she does so indirectly. When sold the DG for \$10, the intermediary is not even making a decision; she had to pay \$10, so she has to keep \$10 in the DG. The first mover has effectively chosen the final outcome—(\$10, \$5, \$0). All that is different is that the first mover is no longer interacting directly with the recipient; now, in between, the intermediary is pressing a button, having no choice other than to click on “Keep \$10.” More than any other pair of scenarios, the intention and responsibility of the first mover is transparently equivalent, yet punishment drops 18 percent.¹⁵

Intermediation also reduces the frequency of punishment. Comparing scenarios with identical outcomes (when the intermediary makes zero profit on top of her \$5 endowment), the likelihood of punishment drops by 15 percent (from 59 percent to 50 percent) when the intermediary is used (see Table A2 in the Appendix). This difference is significant at the 1 percent level using either a Wilcoxon signed rank test or a Fisher Pitman matched-pair permutation test. Moreover, there is a decrease in the frequency of nonzero punishments for every pre-punishment profit level of the first mover other than \$5, and the decrease is statistically significant at at least the 5 percent level for the three highest profit levels, when misbehavior and punishment is highest.

The drop in the frequency of punishment seems to account for most, but not all, of the decrease in the level of punishment. Table 9 in the online Appendix runs the

¹⁵This comparison also answers the hypothesis that punishment is decreasing in response to average overall behavior by the group—Player A is being selfish, but perhaps if Player B is acting nice, the average “niceness” of the group has increased, so I, as a punisher, may act nice as well. See Claudia Keser and Frans van Winden (2000); Fischbacher, Simon Gächter, and Fehr (2001); Bruno S. Frey and Stephan Meier (2004) for evidence of subjects’ behavior being well-predicted by group behavior.

TABLE 2—INTERMEDIATION GAME: SUBJECT-LEVEL RANDOM EFFECTS REGRESSIONS

Dependent variable	OLS	OLS	Ordered probit ^{a,b}	Tobit ^c	Tobit ^d	Probit ^b
	Punishment (\$)	Punishment (\$)	Punishment (\$)	Punishment (\$)	Punishment % of total	Punish {0, 1}
Scenarios included	All (I)	Intermediary makes \$0 (II)	All (III)	All (IV)	All (V)	All (VI)
First mover's profit (\$) over \$5	0.90*** (0.04)	0.90*** (0.05)	0.82*** (0.04)	1.51*** (0.07)	0.19*** (0.01)	0.87*** (0.08)
First mover's profit over \$5 × sold	−0.26*** (0.05)	−0.18*** (0.07)	−0.17*** (0.05)	−0.30*** (0.08)	−0.05*** (0.02)	−0.20** (0.03)
Intermediary's profit	0.05* (0.03)	—	0.07** (0.03)	0.14*** (0.05)	0.03*** (0.01)	0.10** (0.04)
Sold dummy	0.31** (0.15)	0.16 (0.20)	0.15 (0.16)	0.21 (0.27)	0.03 (0.05)	−0.14 (0.23)
Period	−0.28 (0.24)	−0.29 (0.26)	−0.10*** (0.03)	−0.36 (0.52)	−0.09 (0.12)	−0.23 (0.34)
Constant	1.12** (0.21)	1.14 (0.72)	—	−1.99 (1.44)	−0.13 (0.34)	−1.00 (0.98)

Notes: Sixty-four subjects made punishment decisions for 27 scenarios. Model II includes only 12 of these scenarios.
^aOnly integer punishments (11 values) were chosen by subjects.
^bMarginal effects reported.
^cData censored below at \$0.
^dData censored below at 0 percent and above at 100 percent.
***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

same nonparametric analysis as Table 1 but only includes matched pairs where both data are nonzero. There is little evidence of a substantial decrease in punishment due to intermediation, though the decrease remains when the first mover keeps \$10 for herself (also see Model I of Table 10 in the online Appendix for an OLS regression suggesting the decrease remains after controlling for punishment frequency). Carpenter and Matthews (2009) suggest subjects often use a different set of norms or rules when making the decision whether to punish and how much to punish. In this game, intermediation affects both decisions.

Various regression specifications confirm result 1. Table 2 regresses punishment on monetary outcomes and paths taken (“Sold” is a dummy for whether the DG was sold to the intermediary). Every regression is a subject-level random effects model. According to the point estimate in the first row in Model I of Table 2, for every dollar the first mover keeps over \$5 playing the Dictator Game herself, punishment increases 90 cents. The coefficient on the interaction term in the second row is the linear estimate of how much this punishment changes when she sells the DG to the intermediary: She is punished 26 cents less for every dollar she keeps by “selling” the DG, a 29 percent reduction.

Though many of the models in Table 2 impose functional forms, they have many nice features. First, they give an idea of the magnitudes. Second, they allow for inclusion of scenarios where the intermediary makes positive profits in *The Intermediation Game*, which is more difficult in pairwise nonparametric tests.

This is important since punishers exhibit “outcome bias”.¹⁶ Punishment increases slightly, the more the intermediary keeps in the DG. This can be seen on the coefficient on “Intermediary’s Profit” in Table 2. Though only a point estimate, the reported magnitude of the outcome bias, however, is quite small (\$0.05), less than one-fifth the expected decrease in punishment for each dollar made when intermediating (\$0.26). Moreover, if the game is sold for a price near \$10, there is little profit for the intermediary to make, and thus little opportunity for outcome bias to affect punishment.

None of the predictions from Section II are consistent with the results we find in this context. In Section II, we made predictions for what punishments would be for scenarios which differ only in the path taken to the outcome. Table 1 makes just such a comparison. Recall, none of the predictions produced by the fairness models in Section II are consistent with punishment being greater for direct actions. However, Table 1 shows that subjects on average punish direct actions more harshly.¹⁷ Further, Model II in Table 2 reruns the OLS specification in Model I, this time only including scenarios where the intermediary made \$0 profit. Though these scenarios are included in Model I, imposing linearity on the relationship between the intermediary’s profit and punishment when it is not a linear relationship may bias the other estimates in the model. The coefficient on the interaction term in the second row of Model II is also significantly less than 0; thus, even when the intermediary is *completely blameless*, her inclusion reduces punishment. No model discussed in Section II is consistent with this result.

Models III through V in Table 2 also estimate that intermediation significantly reduces punishment. Though punishment could be any whole cent amount, only whole dollar amounts were reported by subjects, so the data only take on eleven values. Additionally, punishment was censored in the game in two ways. First, subjects were not allowed to reward, so punishment was censored below at zero. Second, subjects could not reduce the first mover’s payout to negative amounts, so it was censored above by the first mover’s pre-punishment payout. These discreteness and censoring issues are addressed in Models III through V, using random effects ordered probit and tobit regressions. The interaction term in every one of these regressions is significantly less than zero, consistent with the OLS regressions and the non-parametric analysis. Model VI is a probit regression with a dependent variable that is an indicator of whether there was any punishment or not. The interaction term in this model, as well, is significantly less than zero corroborating the nonparametric analysis that intermediation reduces the likelihood of punishment.¹⁸

Also worth noting is that punishment is increasing in selfishness; Despite punishment coming at no financial cost to themselves, subjects are not just punishing

¹⁶ See Francesca Gino, Don A. Moore, and Bazerman (2008) and Fiery Cushman et al. (2009) for clean demonstrations of outcome bias.

¹⁷ Table 8 in the online Appendix furthers this point by showing how many subjects punish the indirect action harsher.

¹⁸ See online Appendix Table 10 for alternative specifications including session random effects, subject-level and session-level clustering, and subject-level and session-level fixed effects models.

TABLE 3—OLS: TWO PUNISHMENTS

Dep. var. = punishment of A (\$)	Treatment = two punishments		
	N=36	N=24	N=36
	All punishers, punishments (I)	Punishers who never diffuse (II)	Punishments with no diffusion (III)
First mover's profit (\$) over \$5	0.68*** (0.14)	0.67*** (0.16)	0.68*** (0.30)
First mover's profit over \$5 *sold	-0.19** (0.09)	-0.21** (0.08)	-0.30*** (0.08)
Intermediary's profit	0.01 (0.01)	0.00 (0.00)	-0.07 (0.05)
Sold dummy	0.32 (0.20)	0.26*** (0.08)	0.20 (0.12)
Period	0.32** (0.15)	0.26 (0.19)	0.17 (0.16)
Constant	-1.00** (0.39)	-1.05* (0.52)	-0.67 (0.40)

Note: Standard errors clustered at subject level.
***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

noisily. The financial consequences on their lab mates seem to be sufficient incentive for thoughtfulness in their decision-making.

RESULT 2:

Blame is not just being shifted or diffused; it is merely decreasing.

In addition to considering scenarios where the intermediaries are *completely blameless*, we can further our case by looking at these scenarios when zero blame is *actually* assigned to the intermediaries. In the *Two Punishments* treatment, both the first mover and the intermediary could be punished. Thus we can rerun our analyses controlling for whether punishment was shifted or diffused to the intermediary. Models II and III in Table 3 show OLS regression output for such scenarios. These specifications only include data from scenarios where the intermediary made zero profit. Model II controls for punishment-shifting at the subject level; it only includes data from punishers who *never* punished the intermediary when the intermediary made zero profit. Model III controls for punishment-shifting at the punishment level; it only includes punishments of the first mover if the punishment of the intermediary was \$0. For both of these restricted samples, punishment still decreases when the intermediary is used: The coefficient on the first mover's profit interacted with a dummy for selling the game is significantly negative in both cases. Intermediation does not reduce punishment simply by shifting blame onto another player.

RESULT 3:

The availability of intermediation makes the poorest player dramatically worse off and reduces total surplus.

TABLE 4—WELFARE AND EQUALITY

	Intermediation game	No intermediary treatment	Rank sum <i>p</i> -value
Receiver's average payoff	\$1.63	\$4.50	< 0.001
Percent who receive \$5	22 percent	79 percent	
Percent who receive \$0	55 percent	0 percent	
Average profit (all roles)	\$4.31	\$4.85	0.01

As Table 4 indicates, the receiver earns \$1.63 on average in *The Intermediation Game* versus \$4.50 in the *No Intermediary* treatment, 2.8 times more.¹⁹ These data are all earnings from both games and thus are free from selection issues. The only difference is that in one game, there is the option of an intermediary. In the *Third Party Punishment Game* (similar to the *No Intermediary Treatment*), Fehr and Fischbacher (2004) conclude that punishment is successful in encouraging significantly more equitable outcomes. When an intermediary is added to the same game, this is far from being the case. In the presence of an intermediary, punishment has almost completely lost its power to encourage pro-social behavior.

Not only is the availability of an intermediary bad for the poorest player, in this particular setting, it is surplus-damaging. Even though punishment is less when the intermediary is used, it is still greater than zero (The sum of the first two coefficients in Table 2, \$0.64, which is the amount of punishment per dollar kept when selling the DG, is significantly greater than \$0). Since punishment is inefficient, this creates a loss in overall social welfare, assuming the researcher's budget is not a part of the reference group for the social welfare calculation. Finally, since selfishness was much more common in the presence of an intermediary, there was much more punishment, creating substantial surplus losses; group earnings dropped from \$19.55 to \$17.24 when an intermediary is added.

RESULT 4:

Even when the DG is unsold in *The Intermediation Game*, the receiver is worse off than had there not been an intermediary.

Perhaps surprisingly, dictators who *do not* sell the DG keep more than dictators in the *No Intermediation* treatment; they only send \$3.13, significantly less than the \$4.50 sent in the *No Intermediary* treatment (Wilcoxon two-tailed rank sum *p*-value < 0.01). This could be due to selection: Those who send the *most* in the *No Intermediary* treatment might be now selling the DG to the intermediary.²⁰ It could also be a changing norm: The same person will now choose to keep more because there is an intermediary present even if she does not use the intermediary. We cannot

¹⁹ Average profits reported are for all periods, not just the randomly chosen period for which the subjects were paid.

²⁰ This would imply that the subjects who would choose the most generous allocation directly—those most motivated by factors such as guilt aversion, altruism, or blame-avoidance—are now choosing to intermediate; thus, they believe intermediating attenuates these factors. Support for the “blame-avoidance” motive can be found in both Jason Dana, Daylian M. Cain, and Robyn M. Dawes (2005) and John R. Hamman, Loewenstein, and Roberto A. Weber (2010).

TABLE 5—BELIEFS OF HOW MUCH WILL BE KEPT IN THE DG

	Unsold ^a	Purchase price				
		\$5	\$6	\$7	\$8	\$9
Expected amount kept in \$10 DG	\$5.90	\$8.37	\$8.97	\$9.30	\$9.67	\$9.85
Beliefs of others' expectations of amount kept	\$6.10	\$8.30	\$8.90	\$9.45	\$9.75	\$9.95
Average actual	\$5.50	NA	\$10	\$9.43	\$9.75	\$9.83
Amount kept	N=24	N=0	N=2	N=7	N=8	N=6

Note: ^aBeliefs and averages for Unsold column taken from *No Intermediary* treatment.

separate these stories in this experiment though either is an implication of how inter-mediation changes the game that is worthy of future investigation.

RESULT 5:

Subjects believe intermediation makes the poorest player substantially poorer, and they believe others share this belief.

Subjects correctly predict the extreme inequity that intermediation produces. Recall beliefs of how much the intermediary kept in the DG were elicited in an incen-tivized manner after the game was played but before feedback was given. Beliefs of how much the first mover keeps in the DG were elicited in the *No Intermediary* treatment since this eliminates selection of who sells the game.²¹

Table 5 shows that subjects believe that other subjects expect the intermediary to keep at least 67 percent and 95 percent of the remaining value of the DG. In stark contrast, subjects believe the first mover will send the receiver about \$4 when she is not allowed to sell the game. (The first column of Table 5 is from the *No Intermediary* treatment to control for selection.) Moreover, 46 of 64 subjects report that the intermediary will keep more than she was made to pay for the DG for *every* price, *and* that others believe this as well.²² Wilcoxon rank sum tests confirm sub-jects believe less will be sent to the receiver when the game is sold regardless of the price, and they believe other subjects hold similar beliefs ($p < 0.01$ for both, two-tailed test comparing the unsold belief distribution to the sold belief distribution, also for any given price). Thus punishers believe using the intermediary reduces equity and makes the poorest player much worse off, and they believe the first mover shares this belief; however, they punish her less when she uses the intermediary. This is particularly striking because experimental subjects have been shown to have maximin preferences over payoffs (Charness and Rabin 2002). Punishing interme-diation less runs strictly opposite this preference.

If not beliefs, what is driving subjects' behavior? One hypothesis is that some subjects exhibit limited reasoning: Pinning responsibility on the first mover in the case of intermediation takes more reasoning. If some subjects do not reason through the extra logic required, this could drive Result 1. *The Reflection Treatment* was

²¹ Beliefs of how much punishment would occur in each scenario was not elicited. Since punishment is surplus-damaging, taking an action that is expected to be met with punishment might be deemed more unfair. This is an interesting hypothesis for future research.
²² See online Appendix, Table 12 for OLS regressions of these 46 subjects.

designed to test this hypothesis. Recall, subjects were given four neutral questions as primes for reflection before the game was played. Twenty-two of the 24 subjects wrote something for all four questions. 15 explicitly mentioned in at least one of their answers that the first mover will sell the game to avoid punishment and that selling the DG would likely lead to worse outcomes for the receiver. (The tables refer to these 15 as “wise” subjects.)

RESULT 6:

Limited reasoning is not driving Result 1.

Comparing scenarios with identical outcomes, the data indicate even for these 15 wise subjects, intermediation decreases punishment ($p < 0.01$ for both matched pair signed rank test and Fisher Pitman matched-pair permutation test).²³ The wise subjects show a decrease in punishment when the DG is sold in a linear approximation as well; they show an estimated 37 percent decrease in punishment for money made through intermediation.²⁴

Thus even subjects who wrote down that first movers sell the game to try to avoid punishment and that selling is bad for the poorest player, punish first movers less when they sell the game. They exhibit an awareness of the dubious intentions of a first mover who sells as well as the consequences of their behavior, but this does not predict their punishment. Coupling this with Result 5 (subjects believe intermediation makes the poorest player substantially poorer), it has been shown that subjects who fully understand the poor outcomes and dubious intentions involved in intermediation still punish it less; thus, limited rationality does not seem to explain Result 1.

RESULT 7:

The inclusion of a third, endowed player alone does not reduce punishment.

If intermediation reduces punishment because intermediation includes a third party with an endowment, which, among other reasons, may change the reference group for fairness calculations, then we should note a reduction in punishment in *The Allow-Taking Game* when the intermediary is included.

Merely including an endowed third party does not reduce punishment. Subjects punish the first mover fairly equally whether she decides to allow taking or not.²⁵ The data only show one possible significant difference in punishment, and the difference is in the opposite direction: The first mover is punished slightly more for allowing taking after she has kept \$5. The mechanism driving the reduction in punishment by intermediation is not that intermediation includes an endowed third party. It must be another feature of intermediation.

RESULT 8:

Intermediation is not reducing punishment in *The Intermediation Game* wholly due to the market transaction or giving-taking framing effects.

²³ See Table 13 in the online Appendix for average punishments for the “wise” subjects.

²⁴ See Table 14 in the online Appendix for details.

²⁵ Table 3 in the online Appendix shows how subjects punished in *The Allow-Taking Game* based on whether the first mover decided to allow the intermediary to take any of the money sent to the receiver or not.

The Forced-Taking Game was employed to test alternative reasons why intermediation reduces punishment in the baseline but not *The Allow-Taking Game*: Market-framing and giving versus taking framing. If intermediation reduces punishment in *The Forced-Taking Game*, then these explanations are not fully driving Result 1. The data from *The Forced-Taking Game* provide evidence that intermediation reduces punishment in this environment. As in *The Intermediation Game*, this holds when misbehavior, and punishment, is high. When the first mover makes \$10, she is punished less when the intermediary passes her the money than if she took it herself (see Table 4 in the online Appendix for details). The results for \$9 are slightly mixed: the p -value from a matched pair signed rank test is 0.2 while the p -value from a Fisher-Pitman permutation test of the data is 0.07. This indicates that, if there is an effect, it may be driven by few, large differences rather than many, small differences. The effect intermediation has on punishment seems to be smaller, or at least noisier, in this game compared to *The Intermediation Game*. This could be due to, among other reasons, subject pool differences, different sample sizes, and though speculative, framing the use of an intermediary as a market transaction, or switching from a giving frame to a taking frame.

IV. Study 2—Intermediation Reduces Reward

A follow-up study was conducted with two main objectives—first, primarily to test the generalizability of Result 1 and second, to attempt to understand the mechanism driving Result 1. The second experiment was a modified *Intermediation Game* with charitable behavior (real donations to a real charity) and rewards rather than antisocial behavior and punishment.

Subjects were undergraduates at Harvard University. All were recruited via e-mail in November 2008. Each dormitory was randomized into one of two experiments, either the *Donation Experiment* or the *Reward Experiment*, and sent a link for that experiment. The website they visited then randomized them into a treatment, so randomization within each experiment was at the individual level.²⁶ The *Reward Experiment* will provide the data of interest while the *Donation Experiment* will simply provide behaviors for subjects to reward.

A. Donation Experiment Design

Subjects in the first experiment were told that three names would be drawn for every 100 survey respondents. The first name drawn would win \$70 and have the opportunity to donate money to purchase mosquito nets for pregnant mothers in Busia, Kenya.²⁷ This opportunity was framed as either a direct gift to the pregnant mothers or as a donation to a charity (TamTam, www.tamtamafrika.org) who would then purchase nets for pregnant mothers (see the online Appendix, “Charity Experiment Framing” for exact wording). It was made clear that in both cases, a donation of \$3.50 would result in a pregnant mother in Busia receiving a mosquito net and that the

²⁶ Experiment was conducted using Qualtrics survey platform; see www.qualtrics.com.

²⁷ Mosquito nets are very important in this region to protect against malaria.

money would be sent to TamTam, who would deliver the net. The framing manipulated the saliency of passing the donation through the intermediary. Subjects reported the number of nets they would like to donate, up to 20. Subjects were assured they could receive a forwarded e-mail from TamTam if they chose to donate. This could ease any suspicions the money would not be sent while maintaining the feature that giving in this experiment is not to receive approbation from the receiver. The sender is anonymous, so her only motivations are altruism and self-signaling (e.g., Roland Bénabou and Jean Tirole 2006 and Zachary Grossman 2009).

The second and third names drawn would play the two roles of a Dictator Game worth \$48 with efficient giving. The amount sent was scaled up by a factor of three (to encourage more nonzero gifts). Everyone made both a mosquito net donation decision and a dictator decision. The dictator decision always followed their donation decision. Each question was on a separate screen, and at any point in the experiment, they did not know what would be on the next screen. They were not made aware of the second experiment.

B. Donation Experiment Results

The framing manipulation had little to no effect on donations.²⁸ The 94 subjects in the direct frame on average donated 10.7 nets while the 108 in the indirect frame donated 10.2. While their generosity is noteworthy (donating more than half of the \$70 prize), the manipulation is not (Wilcoxon ranksum testing equal distributions: $p = 0.61$).²⁹

There is a significant positive correlation in donating nets and giving in the DG (correlation coefficient = 0.25, $p < 0.01$). Of the 85 subjects who gave nothing in the DG, 73 percent donated at least one mosquito net. This jumps to 97 percent for the 120 subjects who gave *something* in the DG. The DG question was primarily added to incentivize a belief-elicitation in the second experiment. However, it may be noteworthy that the actions are correlated.³⁰

C. Reward Experiment Design

Subjects in this experiment were randomized into either the *reward* treatment or the *DG Guess* treatment. In either experiment, they were then told they would be

²⁸To see sample composition and evidence of randomization, see the online Appendix. Subjects were 97 percent non-seniors, 69 percent women, and 59 percent participants from a popular student charity organization (PBHA). This is not representative of the undergraduate population, so there was sorting; however, even though this may be expected to affect the amount of giving, ex ante, we have no reason to expect this may interact with the treatment.

²⁹Though there was no ex ante hypothesis for this result, it is consistent with the self-serving bias literature. Performing an act with an intermediary may result in ambiguity of who is responsible. Hamman, Loewenstein, and Weber (2010) show that subjects are more likely to undertake *unethical* endeavors with an intermediary, in such cases resolving the ambiguity by convincing themselves the intermediary is responsible. Subjects will resolve this ambiguity in a self-serving way; thus, if responsibility is good, as it is here, they will take full credit, and we should not see any difference between the two conditions.

³⁰This is perhaps most interesting because one decision was made in a natural context—donating money to a charity—while the other was made in a more abstract, artificial context—efficient giving to an anonymous other.

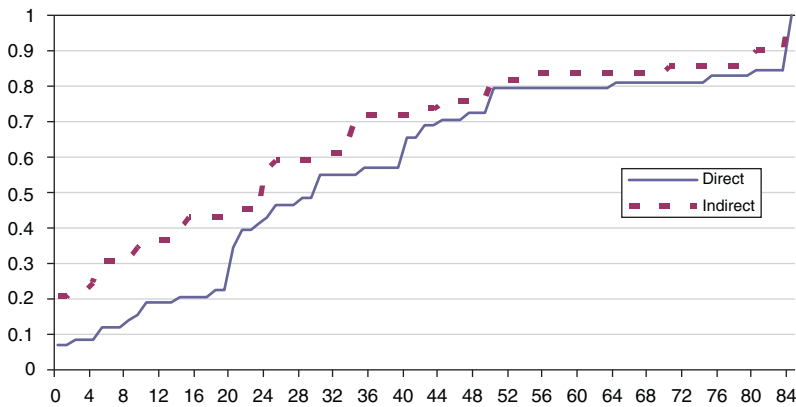


FIGURE 2. CDFs OF REWARDS TO MOSQUITO NET DONORS, BY FRAMING MANIPULATION

randomly matched with one subject from Survey 1. They were told one piece of information about this person. They were (correctly) told either:

Direct Framing.—“This respondent was asked if he/she won \$70, would he/she like to purchase mosquito nets (at \$3.50 each) for pregnant women in Busia, Kenya (through www.tamtamafrica.org). This respondent said yes—had his/her name been drawn he/she would have purchased mosquito nets for pregnant women in Busia, Kenya.”

Indirect Framing.—“This respondent was asked if he/she won \$70, would he/she like to donate money to Tam Tam (www.tamtamafrica.org). Money donated to TamTam enables them to purchase mosquito nets (at \$3.50 each) for pregnant women in Busia, Kenya. This respondent said yes—had his/her name been drawn he/she would have donated money to TamTam.”

Subjects in the *reward* experiment then played a dictator game worth \$84, with efficient giving—so that for each \$1 sent, the other participant received \$3. In the DG, the recipient is the anonymous Survey 1 respondent about whom they just learned. They were told the Survey 1 respondent would receive a letter, along with any money sent, explaining that another respondent was given the opportunity to send them money based on their decision to donate nets. Thus, this was framed as an unexpected reward. They received the letter even in the case of zero money sent.³¹

Subjects in the *DG Guess* experiment did not reward but rather were asked to guess how much the anonymous Survey 1 respondent sent in the DG she played

³¹ This design is most similar to Experiment 2 in Kahneman, Knetsch, and Thaler (1986a), where subjects play as a dictator in a DG with subjects who just finished playing a DG as a dictator and were either entirely selfish or unselfish but did not have their payouts realized, due to a lottery. Receivers who tried to be unselfish as dictators received significantly more than those who had chosen selfish allocations.

following her donation decision. They were paid \$50 if they were within \$5 of how much the respondent actually sent.³²

D. Reward Experiment Results

Subjects rewarded direct-framed giving more than indirect-framed giving. Subjects sent \$36.4 to “direct” donors (thus they received \$109.2), while subjects sent “indirect” donors an average of \$29 (thus they received \$87). With the ex ante hypothesis we would replicate the punishment results—that indirectness decreases rewards—we employ a one-tailed Wilcoxon rank sum, which yields a $p = 0.05$. The cumulative distributions of rewards shown in Figure 2 further show the difference in rewards. Rewards to the direct donor appear to first order stochastically dominate rewards to the indirect donor, and a Kolmogorov-Smirnov test of minimum distance marginally rejects equal probability distributions with a $p = 0.09$.

This is preliminary evidence that the decision process driving Result 1 from *The Intermediation Game* may be generalizable. We have changed the behavior from selfish to charitable, the first mover’s motivation from intrinsic and extrinsic to intrinsic (the first mover knew of the punisher in *The Intermediation Game*, but not of the rewarder here), the environment from the lab to online, from anonymous, undergraduate subject receivers to real charities and charitable cases, and from endogenous, chosen intermediation to exogenously imposed intermediation. In this very different environment, a similar result develops: Intermediation reduces reward as it did punishment.

Finally, subjects in the *DG Guess* experiment believe subjects who donated nets *indirectly* donated *more* in the subsequent DG. They guessed indirect donors sent \$14.1 compared to \$8.5 by direct donors. A two-tailed Wilcoxon rank sum reports these distributions are significantly different, $p = 0.04$. If the DG decision in Survey 1 is believed by subjects in Survey 2 to be independent of the decision to donate (it always came after),³³ then perhaps this may be perceived as a measure of baseline altruism. This question was added to test a hypothesis that Result 1 was being driven not by “punishing bad behavior” but rather “punishing bad people” (e.g., David K. Levine 1998). This hypothesis is not supported in Survey 2; rewards are higher for the group with less expected altruism (direct donors).

V. Related Literature

A. The Economics Literature: Punishment and Fairness

The literature has shown punishment can effectively regulate individuals (Fehr and Fischbacher 2004; Fehr and Gächter 2000) as well as companies (Alexander

³² This elicits their belief of the midpoint of the \$10 interval containing the most Survey 1 donors in their distribution of beliefs. Beliefs were elicited this way in order to keep it simple and understandable for the subject.

³³ The DG decision in Survey 1 may very well not be independent. There might be wealth effects, though wealth is only in expectation. Additionally, often after performing a good act, we feel licensed to be less moral in the next period (e.g., Cain, Loewenstein, and Moore 2005). Whether subjects predict such inter-question dependencies, or others, is unknown to the author. In any case, this result should only be regarded with these considerations.

Dyck and Luigi Zingales 2002; Kenneth J. Martin and Randall S. Thomas 2005). Theoretically, punishment increases pro-social behavior, which means that punishment can lead to greater social welfare for the group (Robert Axelrod 1986). These encouraging results are supported by models of punishment. Theories of *how* we punish are typically a reflection of theories of *why* we punish: If we punish to reinforce pro-social behavior, punishment should be a response to, and only to, antisocial behavior. Hence most models of punishment and fairness are models identifying pro-sociality of behavior. They do this through factors such as outcomes, intentions, and procedures. Fehr and Schmidt (1999) and Bolton and Ockenfels (2000) both provide models of outcome-based fairness. Fehr and Schmidt (1999) model this in a two-player framework, where both players have a preference for equitable outcomes, where advantageous inequity comes at less of a utility cost. Bolton and Ockenfels (2000) argue that players have a preference for their own absolute payoff as well as how their payoff compares to the group average. Subjects in Falk, Fehr, and Fischbacher's (2005) experiment punish defectors more than cooperators even when it is more costly to do so. This suggests behavior, not just outcomes, drives reciprocity. Rabin (1993) and Dufwenberg and Kirchsteiger (2004) define fairness in terms of intentions. Intention is comprised both of strategy space—how good is your action relative to your alternatives?—and your beliefs—given what you think I am going to do, how nice are you being? Falk and Fischbacher (2006) combine intentions and fairness in a theoretical framework that Falk, Fehr, and Fischbacher (2008) provide experimental evidence works best among these three classes of models in predicting behavior in simple, laboratory experiments. Charness and Rabin (2002) provide experimental evidence (and a general model in the Appendix) that subjects' fairness judgments and reciprocity decisions are largely dependent on the welfare of the poorest player in the reference group. Bolton, Brandts, and Ockenfels (2005) provide experimental evidence that subjects' perceptions of fairness also depend on ex ante equity, a concept they term "procedural fairness." Our experiment did not speak to procedural fairness, but it was designed to cleanly test outcomes, intentions and the combination. We found that when pitted directly against "directness," these concepts of fairness made poor predictions. The conclusion is not that outcomes and intentions do not matter, but rather that directness does, and it seems to be first order.

Many experiments have investigated how subjects perceive fairness when multiple agents have acted. The main finding is that responsibility can be diffused when a second party is present.³⁴ Charness (2000), Chaim Fershtman and Uri Gneezy (2001), and Bartling and Fischbacher (forthcoming) all demonstrate experimentally that the addition of a second responsible party may alleviate perceived responsibility of the first party. Fershtman and Gneezy (2001) show this is true even though the first mover is incentivizing the second party to act selfishly on her behalf and even when she has the choice of not using the second party at all. The authors provide the

³⁴ This paper follows the custom of assuming punishment, blame, responsibility, and unfairness are positively correlated. As Falk, Fehr, and Fischbacher (2005) conclude, "the desire to harm those who committed unfair acts, seems to be the most important motive behind fairness-driven sanctions." It must be noted, however, that though these are very similar, they are not identical. Cushman (2008) shows punishment is better predicted by outcomes while judgments without consequence are better predicted by the intentions of the actor.

insight that perhaps the behavior they observed is “because it is not [done] directly by the proposer but by a third party,” though they do not provide experimental support for this specific hypothesis. Rather, they show the subjects care about the second party’s payout; the second party is a “hostage” in the negotiation. Bartling and Fischbacher (forthcoming) show similar abated reciprocity even though the first mover is delegating the choice to a second party who has incentive to act selfishly on behalf of both of them. Moreover, Hamman, Loewenstein, and Weber (2010) show guilt might also decrease in such situations. Experimental subjects demand an agent take more for them in a DG than they are willing to take for themselves (since assignment is random, this also means subjects are willing to take more as an agent for another than they are for themselves). The experiment presented here is designed to allow for analysis of scenarios where the intermediary is completely blameless, and completely unpunished, for the outcome. Even in such scenarios, punishments decrease for actions done through an intermediary. This result does not refute responsibility diffusion. Instead, it identifies a separate force acting in tandem, strengthening the claims that the frequency of bad actions may increase when more people are involved, and that one may want to involve more people when taking a bad action.

A few recent experiments have shown that from a first person perspective, direct interaction may be important in fairness calculations. Dana, Cain, and Dawes (2006) show a significant portion of subjects would prefer to take \$9 than have their decision from a \$10 dictator game implemented. When they take the \$9, the recipient does not receive anything, so this decision is at least weakly dominated by two DG allocations, (\$9, \$1) and (\$10, \$0). The subjects seem to not want to feel responsible for making the decision. Similarly, Edward P. Lazear, Ulrike Malmendier, and Weber (2009) show that most subjects would prefer to avoid the opportunity to share. Dana, Weber, and Xi Kuang (2007) show that many dictators do not costlessly reveal the payout of their recipient, perhaps avoiding having to make a difficult or selfish decision. Stefano DellaVigna, John A. List, and Malmendier (2009) show similar avoidance behavior in the field. When a charity announces they will be coming around the next day for donations, 10–25 percent fewer households answer the door compared to unannounced visits. The results in our paper are psychologically similar, though from a third person perspective. In all of the studies as well as ours, an agent is faced with the decision {Play, Don’t Play} and a sharing decision if she chooses “Play.” Subjects in these experiments behave as if choosing “Don’t Play” is okay even if that means not sharing anything. In our study, selling the DG is similar to choosing “Don’t Play,” and they are judged less harshly for doing so.

B. Moral Psychology

The Moral Psychology literature has shown a much less rational side of punishment. Having subjects rate the morality of a hypothetical scenario, the Moral Psychology literature has identified many ways how and why moral judgment differs from utilitarian standards.³⁵ Kahneman and Shane Frederick (2002) provide

³⁵ See the online Appendix for the most common example, the trolley-footbridge problem.

evidence that punishment and moral judgment are not merely responses to the objective wrongness of an action. Moral preferences, rather, are at least partially derived from an intuitive or emotional reaction see Greene and Jonathan Haidt (2002) for a neuroimaging survey, Haidt (2001) on moral dumbfounding, Madan M. Pillutla and J. Keith Murnighan (1996) on anger in the ultimatum game, Piercarlo Valdesolo and David DeSteno (2006) for exogenous variation in emotion and moral judgments, and Thalia Wheatley and Haidt (2005) on hypnotic disgust and moral judgment; in fact, subjects often are unable to explain why they made the moral judgment that they did (Cushman, Liane Young, and Marc Hauser 2006). This has been explained by a dual process model of the brain, the emotional and the reasoning (see Greene 2007 for review).

The Moral Psychology literature also speaks more specifically to the question of directness and morality. Much research has provided evidence suggesting that blame increases, the more directly the consequence follows from the action. For example, using your “personal force” to kill a man is much worse than pulling a lever that is sure to kill him (Greene et al. 2009). Subjects also claim they would be more willing to undertake actions whose bad outcomes are side effects rather than means to an end (Edward B. Royzman and Jonathan Baron 2002). Pahariya et al. (2009) also show that subjects who read hypothetical scenarios rate firms as less immoral if the firms act selfishly when outsourcing. Interestingly, they find the effect diminishes if they read the two scenarios—when the firm outsources and when they do not—side by side. We add to this literature in three ways. First, we are able to investigate strategic behavior. Since the subjects are playing a game, they are responding to the beliefs and strategies of others. Second, in this environment, we can design clean predictions for existing theories of fairness and understand how they may be improved. Third, subjects are incentivized. This does not always have an effect in experiments, but Colin F. Camerer and Robin M. Hogarth (1999) suggest they may affect subject’s incentive to “present” themselves flatteringly. In a context of selfishness and punishment, these effects may be a concern.

Another interesting finding related to directness has been dubbed “omission bias.” Mark Spranca, Elisa Minsk, and Baron (1991) show that subjects judge an act of omission to be less reprehensible than an act of omission with the same intentions, motives, and consequences. Like the subjects in economic experiments who choose to avoid making a sharing decision, these participants seem to judge a person for the actions she takes, not the ones she does not. This is consistent with our finding that subjects are punished less when they choose not to directly interact with the poorest player.

VI. Concluding Remarks

This paper investigates how an action is punished when performed through an intermediary. We employ a simple game, which allows one player (the first mover) the option of undertaking an antisocial action (keeping money at the expense of the poorest player) and the option of whether to do this directly or through an intermediary. An unaffected party has the opportunity to punish the first mover based on her actions. Using the intermediary is much worse for the poorest player (and equity),

and subjects expect this (and believe others expect this). Nevertheless, subjects punish the first mover less when she elects to keep money via intermediation. This is true even if the intermediary is unambiguously free of responsibility. We then perform a treatment that encourages subjects to think and write about the game and the strategies of the other players before they play. Fifteen of 24 write down that the first mover will use the intermediary to avoid punishment *and* that intermediation is worse for the poorest player. When these 15 subjects subsequently play the game, they punish intermediation less. Thus even when subjects understand the poor outcomes intermediation produces or the punishment-skirting intentions of the first mover, they punish in a manner suggesting they believe using the intermediary is actually less punishable. We further show the punishment reduction is not due to the mere inclusion of a third player. Across treatments, punishment decreases if and only if including the intermediary allows the first mover to avoid directly interacting with the poorest player. In a framed field study, we show a similar effect with rewards and charitable action. These reductions in punishment and reward due to directness is interesting for two reasons: first, they cannot be explained by any current model or literature in Economics. Second, intermediation is widespread, and perhaps our moral intuitions and judgments in the case of individual or corporate partnerships should be questioned.

APPENDIX

BEHAVIORAL PREDICTIONS APPENDIX

A. Outcome-Based Models

Fehr and Schmidt (1999) suggest the punisher's utility of payoff vector $\mathbf{x}$ takes the form:

$$(1) \quad U_i(x) = x_i - \alpha_i \frac{1}{n-1} \sum_{j \neq i} \max\{x_j - x_i, 0\} \\ - \beta_i \frac{1}{n-1} \sum_{j \neq i} \max\{x_i - x_j, 0\},$$

where $\beta_i \leq \alpha_i$ and $0 \leq \beta_i \leq 1$. That is, utility is defined by her own payout, x_i , but her utility takes a hit for inequality in payouts, and more so for disadvantageous inequalities. In this model, under either path in the *Intermediation Game*, the punisher's utility is thus (Recall $X \geq 5$):

$$(2) \quad U_i(X, 5, 10 - X, 5) = 5 - \alpha_i \frac{1}{3}(X - 5) - \beta_i \frac{1}{3}(X - 5),$$

which reduces to 5 if $X = 5$ since there is no longer any inequality. Most importantly, no matter the size of X , the estimated utilities are identical, and we have no reason to predict a difference in punishment for using the intermediary.

TABLE A1—SUMMARY STATISTICS

	Intermediation game	No intermediary treatment
Number of subjects	64	24
Number who use intermediary	33	—
Average sell price	\$7.17	—
Average amount kept in \$10 DG	\$8.37	\$5.50
(when sold)	\$9.79	—
(when unsold)	\$6.87	—
Percent who punish at least once	81%	100%
Average punishment	\$2.07	\$3.65
Average nonzero punishment	\$4.24	\$4.70
Average punishment as percentage of first mover's pre-punishment wealth	28%	44%
Average nonzero punishment as percentage of first mover's pre-punishment wealth	58%	56%
Average profit (all subjects)	\$4.31	\$4.83

TABLE A2—DIRECTNESS IS PUNISHED MORE FREQUENTLY^a

Profit of first mover	Percent who punish		Number of subjects who punish		<i>p</i> -values	
	DG unsold	DG sold	Unsold, but not sold	Sold, but not unsold	Matched pair, signed rank ^b	Permutation test ^c
All	59	50	41	8	< 0.01	< 0.01
\$10	75	67	8	3	0.13	0.23
\$9	77	67	6	0	0.01	0.03
\$8	75	63	9	1	0.01	0.02
\$7	66	50	11	1	< 0.01	0.01
\$6	45	38	6	1	0.06	0.12
\$5	17	19	1	2	0.56	1.00

Notes: Pairs matched by subject. *p*-values are from two-tailed tests.
^aTo hold outcomes constant, only scenarios where intermediary makes no money included.
^bMatched Pair Signed Rank test based on null hypothesis that punishment distributions are identical for direct and indirect action. Pairs matched by subject.
^cPermutation *p*-values based on 200,000 simulations per test.

Note that $U_i(x)$ can only decrease under intermediation. If the intermediary makes $\$Y > 0$ profit, instead of \$0 as in our driving example, then utility decreases by the added inequality, both the disadvantageous (the intermediary now has $\$Y$ more than the punisher— $\alpha_i \frac{1}{3}(Y)$)—and the advantageous (the receiver now has $\$Y$ less than the punisher relative to before)— $\beta_i \frac{1}{3}(5 - (10 - X - Y))$. Whom to blame for this decrease in utility is not directly addressed by outcome-based models. For that, we add intentions.

B. Outcomes and Intentions Models

Let us consider one very simple model of preference for maximizing the minimum payoff:

(3)
$$U_i(x) = x_i + \xi \cdot \min_j \{x_j\}$$

where $0 \leq \xi \leq 1$. Under these preferences,

$$(4) \quad U_i(X, 5, 10 - X, 5) = 5 - \xi(10 - X)$$

$$(5) \quad U_i(X, 5 + \hat{Y}, 10 - X - \hat{Y}, 5) = 5 - \xi(10 - X - \hat{Y}).$$

Hence, utility decreases by $\xi \cdot \hat{Y}$ under intermediation, and we would expect weakly more punishment, and strictly more if $\hat{Y} > 0$.

Under Fehr and Schmidt (1999) inequity aversion, utility for not using the intermediary— $U_i(X, 5, 10 - X, 5)$ —is given by equation (2) above. Utility via intermediation is given by:

$$(6) \quad U_i(X, 5 + \hat{Y}, 10 - X - \hat{Y}, 5) = 5 - \alpha_i \frac{1}{3} [(X - 5) + \hat{Y}] \\ - \beta_i \frac{1}{3} (X + \hat{Y} - 5).$$

Equation (2)—equation (6) yields a difference of $\alpha_i \frac{1}{3} \hat{Y} + \beta_i \frac{1}{3} \hat{Y}$; hence, utility weakly decreases under intermediation, and we will expect weakly more punishment. Both statements become strict inequalities if $\hat{Y} > 0$.

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